# Partially Asynchronous Optimization on Manifolds

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# I. Simulations

## A. Background on Unit Quaternions

Let $q \in S^3$ be a unit quaternion, given by

$$q = q_w + q_x i + q_y j + q_z k = q_r + q_v \in S^3.$$

The conjugate of $q$ is denoted $q^*$ and given by

$$q^* = q_w - q_x i - q_y j - q_z k = q_r - q_v.$$

It holds that $(q^*)^* = q$, and $(pq)^* = q^* p^*$.

The quaternion product is given by

$$\begin{aligned} pq &= (p_r q_r - p_v \cdot q_v) + (p_r q_v + q_r p_v + p_v \times q_v) \\ &= (p \cdot q^*) + (p_r q_v + q_r p_v + p_v \times q_v). \end{aligned}$$

A rotation of $\theta$ radians around the axis $u \in \mathbb{R}^3$ is given by

$$q = \cos\frac{\theta}{2} + u \sin\frac{\theta}{2}.$$

Represent a vector $v_v \in \mathbb{R}^3$ as a pure quaternion, $v = (0, v_v)$. The rotation of a vector $v_v \in \mathbb{R}^3$ by a unit quaternion $q$ is given by

$$L_q(v) = qvq^*.$$

It follows that the composition of $L_{q2}$ and $L_{q1}$ is given by

$$L_{q2}(L_{q1}(v)) = L_{q2\,q1}(v).$$

Equivalently, the rotation of a vector by a quaternion can written using rotation matrices. Let $R(q) \in SO(3)$ be the rotation matrix induced by $q$, such that

$$L_q(v) = R(q)v_v,$$

where

$$R(q) = \begin{bmatrix} 1 - 2(q_y^2 + q_z^2) & 2(q_x q_y - q_z q_w) & 2(q_x q_z + q_y q_w) \\ 2(q_x q_y + q_z q_w) & 1 - 2(q_x^2 + q_z^2) & 2(q_y q_z - q_x q_w) \\ 2(q_x q_z - q_y q_w) & 2(q_y q_z + q_x q_w) & 1 - 2(q_x^2 + q_y^2) \end{bmatrix}.$$

This representation can be differentiated with respect to $q_w$, $q_x$, $q_y$, and $q_z$, via

$$\begin{aligned} \frac{\partial R(q)}{\partial q_w} &= \begin{bmatrix} 0 & -2q_z & 2q_y \\ 2q_z & 0 & -2q_x \\ -2q_y & 2q_x & 0 \end{bmatrix}, \\ \frac{\partial R(q)}{\partial q_x} &= \begin{bmatrix} 0 & 2q_y & 2q_z \\ 2q_y & -4q_x & -2q_w \\ 2q_z & 2q_w & -4q_x \end{bmatrix}, \\ \frac{\partial R(q)}{\partial q_y} &= \begin{bmatrix} -4q_y & 2q_x & 2q_w \\ 2q_x & 0 & 2q_z \\ -2q_w & 2q_z & -4q_y \end{bmatrix}, \\ \frac{\partial R(q)}{\partial q_z} &= \begin{bmatrix} -4q_z & -2q_w & 2q_x \\ 2q_w & -4q_z & 2q_y \\ 2q_x & 2q_y & 0 \end{bmatrix}. \end{aligned}$$

Let $f$ be some function of $R(q)$, and let $\nabla_{R(q)} f$ denote the gradient of $f$ with respect to $R(q)$. Then, the gradient of $f$ with respect to $q$ is given by

$$\nabla_q f = \begin{bmatrix} \langle \nabla_{R(q)} f, \frac{\partial R(q)}{\partial q_w} \rangle_F \\ \langle \nabla_{R(q)} f, \frac{\partial R(q)}{\partial q_x} \rangle_F \\ \langle \nabla_{R(q)} f, \frac{\partial R(q)}{\partial q_y} \rangle_F \\ \langle \nabla_{R(q)} f, \frac{\partial R(q)}{\partial q_z} \rangle_F \end{bmatrix}.$$

Let $c \in \{w, x, y, z\}$. For clarity, we can see

$$\begin{aligned}
\left\langle \nabla_{R(q)} f, \frac{\partial R(q)}{\partial q_c} \right\rangle_F &= \left\langle \begin{bmatrix} \frac{\partial f}{\partial R(q)_{11}} & \frac{\partial f}{\partial R(q)_{12}} & \frac{\partial f}{\partial R(q)_{13}} \\ \frac{\partial f}{\partial R(q)_{21}} & \frac{\partial f}{\partial R(q)_{22}} & \frac{\partial f}{\partial R(q)_{23}} \\ \frac{\partial f}{\partial R(q)_{31}} & \frac{\partial f}{\partial R(q)_{32}} & \frac{\partial f}{\partial R(q)_{33}} \end{bmatrix}, \begin{bmatrix} \frac{\partial R(q)_{11}}{\partial q_c} & \frac{\partial R(q)_{12}}{\partial q_c} & \frac{\partial R(q)_{13}}{\partial q_c} \\ \frac{\partial R(q)_{21}}{\partial q_c} & \frac{\partial R(q)_{22}}{\partial q_c} & \frac{\partial R(q)_{33}}{\partial q_c} \\ \frac{\partial R(q)_{31}}{\partial q_c} & \frac{\partial R(q)_{32}}{\partial q_c} & \frac{\partial R(q)_{33}}{\partial q_c} \end{bmatrix} \right\rangle_F \\
&= \mathrm{Tr}\left( \begin{bmatrix} \frac{\partial f}{\partial R(q)_{11}} & \frac{\partial f}{\partial R(q)_{12}} & \frac{\partial f}{\partial R(q)_{13}} \\ \frac{\partial f}{\partial R(q)_{21}} & \frac{\partial f}{\partial R(q)_{22}} & \frac{\partial f}{\partial R(q)_{23}} \\ \frac{\partial f}{\partial R(q)_{31}} & \frac{\partial f}{\partial R(q)_{32}} & \frac{\partial f}{\partial R(q)_{33}} \end{bmatrix}^\top \begin{bmatrix} \frac{\partial R(q)_{11}}{\partial q_c} & \frac{\partial R(q)_{12}}{\partial q_c} & \frac{\partial R(q)_{13}}{\partial q_c} \\ \frac{\partial R(q)_{21}}{\partial q_c} & \frac{\partial R(q)_{22}}{\partial q_c} & \frac{\partial R(q)_{33}}{\partial q_c} \\ \frac{\partial R(q)_{31}}{\partial q_c} & \frac{\partial R(q)_{32}}{\partial q_c} & \frac{\partial R(q)_{33}}{\partial q_c} \end{bmatrix} \right) \\
&= \mathrm{Tr}\left( \begin{bmatrix} \frac{\partial f}{\partial R(q)_{11}} & \frac{\partial f}{\partial R(q)_{21}} & \frac{\partial f}{\partial R(q)_{31}} \\ \frac{\partial f}{\partial R(q)_{12}} & \frac{\partial f}{\partial R(q)_{22}} & \frac{\partial f}{\partial R(q)_{32}} \\ \frac{\partial f}{\partial R(q)_{13}} & \frac{\partial f}{\partial R(q)_{23}} & \frac{\partial f}{\partial R(q)_{33}} \end{bmatrix} \begin{bmatrix} \frac{\partial R(q)_{11}}{\partial q_c} & \frac{\partial R(q)_{12}}{\partial q_c} & \frac{\partial R(q)_{13}}{\partial q_c} \\ \frac{\partial R(q)_{21}}{\partial q_c} & \frac{\partial R(q)_{22}}{\partial q_c} & \frac{\partial R(q)_{33}}{\partial q_c} \\ \frac{\partial R(q)_{31}}{\partial q_c} & \frac{\partial R(q)_{32}}{\partial q_c} & \frac{\partial R(q)_{33}}{\partial q_c} \end{bmatrix} \right) \\
&= \frac{\partial f}{\partial R(q)_{11}} \frac{\partial R(q)_{11}}{\partial q_c} + \frac{\partial f}{\partial R(q)_{21}} \frac{\partial R(q)_{21}}{\partial q_c} + \frac{\partial f}{\partial R(q)_{31}} \frac{\partial R(q)_{31}}{\partial q_c} \\
&\quad + \frac{\partial f}{\partial R(q)_{12}} \frac{\partial R(q)_{12}}{\partial q_c} + \frac{\partial f}{\partial R(q)_{22}} \frac{\partial R(q)_{22}}{\partial q_c} + \frac{\partial f}{\partial R(q)_{32}} \frac{\partial R(q)_{32}}{\partial q_c} \\
&\quad + \frac{\partial f}{\partial R(q)_{13}} \frac{\partial R(q)_{13}}{\partial q_c} + \frac{\partial f}{\partial R(q)_{23}} \frac{\partial R(q)_{23}}{\partial q_c} + \frac{\partial f}{\partial R(q)_{33}} \frac{\partial R(q)_{33}}{\partial q_c} \\
&= \frac{\partial f}{\partial q_c}.
\end{aligned}$$

### *B. Toy Euclidean Problem*

Manifold: 3-dimensional Euclidean space,

$$\mathcal{M}_i = \mathbb{R}^3.$$

Objective function:

$$f(\mathcal{D}) = \sum_{(i,j)\in E} \left\| x_j - x_i - \tilde{d}_{ij} \right\|^2_{R_{ij}^{-1}},$$

where $E \in [N]^2$ is a set of directed measurement edges and $\tilde{d}_{ij}$ is a noisy measurement given by

$$\tilde{d}_{ij} = x_j - x_i + v_{ij},$$

with zero-mean Gaussian measurement noise $v_{ij} \in \mathcal{N}(\mathbf{0}^3, R_{ij})$ and covariance matrix $R_{ij} = R_{ij}^\top \succ 0$.

Gradient:

$$\begin{aligned}
f(\mathcal{D}) &= \sum_{(i,j)\in E} \left\| x_j - x_i - \tilde{d}_{ij} \right\|^2_{R_{ij}^{-1}} \\
&= \sum_{(i,j)\in E} (x_j - x_i - \tilde{d}_{ij})^\top R_{ij}^{-1} (x_j - x_i - \tilde{d}_{ij}).
\end{aligned}$$

The gradient of $f$ with respect to $x_i \in \mathcal{M}_i$ is given by

$$\mathrm{grad}_{x_i} f(x) = -2 \sum_{j:(i,j)\in E} R_{ij}^{-1}(x_j - x_i - \tilde{d}_{ij}) + 2 \sum_{j:(j,i)\in E} R_{ji}^{-1}(x_i - x_j - \tilde{d}_{ji}).$$

### *C. Rotation Averaging*

Manifold: Unit Quaternion,

$$\mathcal{M}_i = S^3.$$

Objective function:

$$f(x) = \sum_{(i,j)\in E} \left\| R(x_i)^\top R(x_j) - \tilde{R}_{ij} \right\|^2_F,$$

where $E \in [N]^2$ is a set of directed measurement edges and $\tilde{R}_{ij}$ is a noisy measurement given by

$$\tilde{R}_{ij} = R(x_i)^\top R(x_j) V_{ij},$$

where $V_{ij}$ is a perturbation rotation matrix corresponding to the rotation about a random axis by a zero-mean Gaussian angle.

Gradient: Let $\bar{f}$ be the smooth extension of $f$ to $\mathbb{R}^4$, given by

$$\bar{f}(x) = \sum_{(i,j)\in E} \left\| R(x_i)^\top R(x_j) - \tilde{R}_{ij} \right\|_F^2.$$

Define $R_i, R_j \in SO(3)$ via

$$R_i = R(x_i)$$
$$R_j = R(x_j).$$

Plug in $R_i$ for all $i \in [N]$ into $\bar{f}$. Note that for all $i \in [N]$, the variable $x_i$ only appears through $R_i$. As such, we define $\tilde{f}$ in terms of $R_i$ via

$$\tilde{f}(\{R_1, \ldots, R_N\}) = \sum_{(i,j)\in E} \left\| R_i^\top R_j - \tilde{R}_{ij} \right\|_F^2.$$

Choose $i \in [N]$ and choose a matrix $V_i \in \mathbb{R}^{3\times 3}$. Define $V = \{\mathbf{0}, \ldots, V_i, \ldots, \mathbf{0}\}$ Then,

$$
\begin{aligned}
D\tilde{f}(\{R_1,\ldots,R_N\})[V] &= \lim_{t\to 0} \frac{\tilde{f}(\{R_1,\ldots,R_i+tV_i,\ldots,R_N\}) - \tilde{f}(\{R_1,\ldots,R_N\})}{t} \\
&= \lim_{t\to 0} \frac{1}{t}\Bigg( \sum_{j:(i,j)\in E} \left( \left\| (R_i+tV_i)^\top R_j - \tilde{R}_{ij} \right\|_F^2 - \left\| R_i^\top R_j - \tilde{R}_{ij} \right\|_F^2 \right) \\
&\quad + \sum_{j:(j,i)\in E} \left( \left\| R_j^\top (R_i+tV_i) - \tilde{R}_{ji} \right\|_F^2 - \left\| R_j^\top R_i - \tilde{R}_{ji} \right\|_F^2 \right) \Bigg) \\
&= \lim_{t\to 0} \frac{1}{t}\Bigg( \sum_{j:(i,j)\in E} \left( \left\| R_i^\top R_j - \tilde{R}_{ij} + tV_i^\top R_j \right\|_F^2 - \left\| R_i^\top R_j - \tilde{R}_{ij} \right\|_F^2 \right) \\
&\quad + \sum_{j:(j,i)\in E} \left( \left\| R_j^\top R_i - \tilde{R}_{ji} + tR_j^\top V_i \right\|_F^2 - \left\| R_j^\top R_i - \tilde{R}_{ji} \right\|_F^2 \right) \Bigg) \\
&= \lim_{t\to 0} \frac{1}{t}\Bigg( \sum_{j:(i,j)\in E} \mathrm{Tr}\left( 2\left(tV_i^\top R_j\right)^\top \left( R_i^\top R_j - \tilde{R}_{ij} \right) + \left(tV_i^\top R_j\right)^\top \left(tV_i^\top R_j\right) \right) \\
&\quad + \sum_{j:(j,i)\in E} \mathrm{Tr}\left( 2\left(tR_j^\top V_i\right)^\top \left( R_j^\top R_i - \tilde{R}_{ji} \right) + \left(tR_j^\top V_i\right)^\top \left(tR_j^\top V_i\right) \right) \Bigg) \\
&= \lim_{t\to 0} \frac{1}{t}\Bigg( \sum_{j:(i,j)\in E} \mathrm{Tr}\left( 2tR_j^\top V_i R_i^\top R_j - 2tR_j^\top V_i \tilde{R}_{ij} + t^2 R_j^\top V_i V_i^\top R_j \right) \\
&\quad + \sum_{j:(j,i)\in E} \mathrm{Tr}\left( 2tV_i^\top R_j R_j^\top R_i - 2tV_i^\top R_j \tilde{R}_{ji} + t^2 V_i^\top R_j R_j^\top V_i \right) \Bigg) \\
&= \lim_{t\to 0} \frac{1}{t}\Bigg( \sum_{j:(i,j)\in E} \mathrm{Tr}\left( 2tV_i R_i^\top - 2tR_j^\top V_i \tilde{R}_{ij} + t^2 V_i V_i^\top \right) \\
&\quad + \sum_{j:(j,i)\in E} \mathrm{Tr}\left( 2tV_i^\top R_i - 2tV_i^\top R_j \tilde{R}_{ji} + t^2 V_i^\top V_i \right) \Bigg) \\
&= \lim_{t\to 0} \Bigg( \sum_{j:(i,j)\in E} \mathrm{Tr}\left( 2V_i R_i^\top - 2R_j^\top V_i \tilde{R}_{ij} + tV_i V_i^\top \right) \\
&\quad + \sum_{j:(j,i)\in E} \mathrm{Tr}\left( 2V_i^\top R_i - 2V_i^\top R_j \tilde{R}_{ji} + tV_i^\top V_i \right) \Bigg) \\
&= \sum_{j:(i,j)\in E} \mathrm{Tr}\left( 2V_i R_i^\top - 2R_j^\top V_i \tilde{R}_{ij} \right) + \sum_{j:(j,i)\in E} \mathrm{Tr}\left( 2V_i^\top R_i - 2V_i^\top R_j \tilde{R}_{ji} \right) \\
&= \mathrm{Tr}\left( \sum_{j:(i,j)\in E} \left( 2V_i R_i^\top - 2R_j^\top V_i \tilde{R}_{ij} \right) + \sum_{j:(j,i)\in E} \left( 2V_i^\top R_i - 2V_i^\top R_j \tilde{R}_{ji} \right) \right) \\
&= \mathrm{Tr}\left( \sum_{j:(i,j)\in E} \left( 2V_i^\top R_i - 2V_i^\top R_j \tilde{R}_{ij}^\top \right) + \sum_{j:(j,i)\in E} \left( 2V_i^\top R_i - 2V_i^\top R_j \tilde{R}_{ji} \right) \right) \\
&= \mathrm{Tr}\left( V_i^\top \left( \sum_{j:(i,j)\in E} \left( 2R_i - 2R_j \tilde{R}_{ij}^\top \right) + \sum_{j:(j,i)\in E} \left( 2R_i - 2R_j \tilde{R}_{ji} \right) \right) \right) \\
&= \mathrm{Tr}\left( V_i^\top \nabla_{R_i} \tilde{f} \right)
\end{aligned}
$$

Thus, we know

$$
\nabla_{R_i} \tilde{f}(\{R_1,\ldots,R_N\}) = \sum_{j:(i,j)\in E} \left( 2R_i - 2R_j \tilde{R}_{ij}^\top \right) + \sum_{j:(j,i)\in E} \left( 2R_i - 2R_j \tilde{R}_{ji} \right).
$$

Equivalently, we know

$$\nabla_{R(x_i)}\bar{f}(x) = \sum_{j:(i,j)\in E}\left(2R(x_i) - 2R(x_j)\tilde{R}_{ij}^\top\right) + \sum_{j:(j,i)\in E}\left(2R(x_i) - 2R(x_j)\tilde{R}_{ji}\right).$$

Finally, we again use chain rule, according to

$$\nabla_{x_i}\bar{f}(x) = \begin{bmatrix} \langle \nabla_{R(x_i)}\bar{f}, \frac{\partial R(x_i)}{\partial q_w}\rangle_F \\ \langle \nabla_{R(x_i)}\bar{f}, \frac{\partial R(x_i)}{\partial q_x}\rangle_F \\ \langle \nabla_{R(x_i)}\bar{f}, \frac{\partial R(x_i)}{\partial q_y}\rangle_F \\ \langle \nabla_{R(x_i)}\bar{f}, \frac{\partial R(x_i)}{\partial q_z}\rangle_F \end{bmatrix}$$

with

$$\text{grad}_{x_i} f = P_{x_i}(\nabla_{x_i}\bar{f}).$$

### *D. Rotation-Invariant Formation Control*

Agent $i$'s manifold is given by

$$\mathcal{M}_i = \mathbb{R}^3 \times S^3.$$

Relevant variables for agent $i$:

- Decision variable: $x_i = (t_i, q_i) \in (\mathbb{R}^3 \times S^3) \in \mathcal{M}_i$
- Prior position: $\bar{t}_i \in \mathbb{R}^3$
- Prior orientation: $\bar{R}_i \in SO(3)$
- Relative offset to agent $j$ in local reference frame: $\tilde{t}_{ij}$

Objective function:

$$f(x) = \alpha\sum_{i=1}^{N}\left(\|t_i - \bar{t}_i\|_2^2 + \left\|R(q_i) - \bar{R}_i\right\|_F^2\right) + (1-\alpha)\sum_{(i,j)\in E}\left\|R(q_i)^\top(t_j - t_i) - \tilde{t}_{ij}\right\|_2^2.$$

Gradient with respect to $t_i$:

$$\begin{aligned} f(x) &= \alpha\sum_{i=1}^{N}\left(\|t_i - \bar{t}_i\|_2^2 + \left\|R(q_i) - \bar{R}_i\right\|_F^2\right) + (1-\alpha)\sum_{(i,j)\in E}\left\|R(q_i)^\top(t_j - t_i) - \tilde{t}_{ij}\right\|_2^2 \\ &= \alpha\sum_{i=1}^{N}\left(t_i^\top t_i - 2t_i^\top\bar{t}_i + \bar{t}_i^\top\bar{t}_i + \left\|R(q_i) - \bar{R}_i\right\|_F^2\right) \\ &\quad + (1-\alpha)\sum_{(i,j)\in E}\left((t_j - t_i)^\top R(q_i)R(q_i)^\top(t_j - t_i) - 2(t_j - t_i)^\top R(q_i)\tilde{t}_{ij} + \tilde{t}_{ij}^\top\tilde{t}_{ij}\right) \\ &= \alpha\sum_{i=1}^{N}\left(t_i^\top t_i - 2t_i^\top\bar{t}_i + \bar{t}_i^\top\bar{t}_i + \left\|R(q_i) - \bar{R}_i\right\|_F^2\right) \\ &\quad + (1-\alpha)\sum_{(i,j)\in E}\left((t_j - t_i)^\top(t_j - t_i) - 2(t_j - t_i)^\top R(q_i)\tilde{t}_{ij} + \tilde{t}_{ij}^\top\tilde{t}_{ij}\right) \\ &= \alpha\sum_{i=1}^{N}\left(t_i^\top t_i - 2t_i^\top\bar{t}_i + \bar{t}_i^\top\bar{t}_i + \left\|R(q_i) - \bar{R}_i\right\|_F^2\right) \\ &\quad + (1-\alpha)\sum_{(i,j)\in E}\left(t_j^\top t_j - 2t_i^\top t_j + t_i^\top t_i - 2(t_j - t_i)^\top R(q_i)\tilde{t}_{ij} + \tilde{t}_{ij}^\top\tilde{t}_{ij}\right) \end{aligned}$$

$$\text{grad}_{t_i} f(x) = 2\alpha(t_i - \bar{t}_i) + 2(1-\alpha)\sum_{j:(i,j)\in E}(t_i - t_j + R(q_i)\tilde{t}_{ij}) + 2(1-\alpha)\sum_{j:(j,i)\in E}(t_i - t_j - R(q_j)\tilde{t}_{ji})$$

Gradient with respect to $q_i$:

First, define $\bar{f}$ as smooth extension of $f$, where $\mathcal{M}_i$ is extended to $(\mathbb{R}^3 \times \mathbb{R}^4)$. Additionally, define $R_i = R(q_i)$ and define $\tilde{f}$ as $\bar{f}$ with all instances of $R(q_i)$ replaced with $R_i$, given by

$$\tilde{f}(\{t_i, R_i\}_{i\in[N]}) = \alpha\sum_{i=1}^{N}\left(\|t_i - \bar{t}_i\|_2^2 + \left\|R_i - \bar{R}_i\right\|_F^2\right) + (1-\alpha)\sum_{(i,j)\in E}\left\|R_i^\top(t_j - t_i) - \tilde{t}_{ij}\right\|_2^2.$$

Choose a matrix $V_i \in \mathbb{R}^{3\times 3}$, and define $V = \{(\mathbf{0}^3, \mathbf{0}^{3\times 3})^{i-1}, (\mathbf{0}, V_i), (\mathbf{0}^3, \mathbf{0}^{3\times 3})^{N-i}\}$. Then,

$$
\begin{aligned}
D\tilde{f}(\{t_i, R_i\}_{i\in[N]})[V] &= \lim_{t\to 0} \frac{\tilde{f}((t_1, R_1), \ldots, (t_i, R_i + tV_i), \ldots, (t_N, R_N)) - \tilde{f}(\{t_i, R_i\}_{i\in[N]})}{t} \\
&= \lim_{t\to 0} \frac{1}{t}\Bigg(\alpha \left\|R_i + tV_i - \bar{R}_i\right\|_F^2 - \alpha \left\|R_i - \bar{R}_i\right\|_F^2 \\
&\quad + (1-\alpha) \sum_{j:(i,j)\in E} \left(\left\|(R_i + tV_i)^\top (t_j - t_i) - \tilde{t}_{ij}\right\|_2^2 - \left\|R_i^\top (t_j - t_i) - \tilde{t}_{ij}\right\|_2^2\right)\Bigg) \\
&= \lim_{t\to 0} \frac{1}{t}\Bigg(\alpha \left\|R_i - \bar{R}_i + tV_i\right\|_F^2 - \alpha \left\|R_i - \bar{R}_i\right\|_F^2 \\
&\quad + (1-\alpha) \sum_{j:(i,j)\in E} \left(\left\|R_i^\top (t_j - t_i) - \tilde{t}_{ij} + tV_i^\top (t_j - t_i)\right\|_2^2 - \left\|R_i^\top (t_j - t_i) - \tilde{t}_{ij}\right\|_2^2\right)\Bigg) \\
&= \lim_{t\to 0} \frac{1}{t}\Bigg(\alpha \mathrm{Tr}\left(2tV_i^\top (R_i - \bar{R}_i) + t^2 V_i^\top V_i\right) \\
&\quad + (1-\alpha) \sum_{j:(i,j)\in E} \left(2t(R_i^\top (t_j - t_i) - \tilde{t}_{ij})^\top V_i^\top (t_j - t_i) + t^2 (t_j - t_i)^\top V_i V_i^\top (t_j - t_i)\right)\Bigg) \\
&= \lim_{t\to 0} \Bigg(\alpha \mathrm{Tr}\left(2V_i^\top (R_i - \bar{R}_i) + tV_i^\top V_i\right) \\
&\quad + (1-\alpha) \sum_{j:(i,j)\in E} \left(2(R_i^\top (t_j - t_i) - \tilde{t}_{ij})^\top V_i^\top (t_j - t_i) + t(t_j - t_i)^\top V_i V_i^\top (t_j - t_i)\right)\Bigg) \\
&= \alpha \mathrm{Tr}\left(2V_i^\top (R_i - \bar{R}_i)\right) + (1-\alpha) \sum_{j:(i,j)\in E} 2(R_i^\top (t_j - t_i) - \tilde{t}_{ij})^\top V_i^\top (t_j - t_i) \\
&= \alpha \mathrm{Tr}\left(2V_i^\top (R_i - \bar{R}_i)\right) + (1-\alpha) \mathrm{Tr}\left(\sum_{j:(i,j)\in E} 2(R_i^\top (t_j - t_i) - \tilde{t}_{ij})^\top V_i^\top (t_j - t_i)\right) \\
&= \alpha \mathrm{Tr}\left(2V_i^\top (R_i - \bar{R}_i)\right) + (1-\alpha) \mathrm{Tr}\left(\sum_{j:(i,j)\in E} 2V_i^\top (t_j - t_i)(R_i^\top (t_j - t_i) - \tilde{t}_{ij})^\top\right) \\
&= \mathrm{Tr}\left(2\alpha V_i^\top (R_i - \bar{R}_i) + (1-\alpha) \sum_{j:(i,j)\in E} 2V_i^\top (t_j - t_i)(R_i^\top (t_j - t_i) - \tilde{t}_{ij})^\top\right) \\
&= \mathrm{Tr}\left(V_i^\top \left(2\alpha (R_i - \bar{R}_i) + (1-\alpha) \sum_{j:(i,j)\in E} 2(t_j - t_i)(R_i^\top (t_j - t_i) - \tilde{t}_{ij})^\top\right)\right) \\
&= \mathrm{Tr}\left(V_i^\top \nabla_{R_i} \tilde{f}\right)
\end{aligned}
$$

Thus, we know

$$\nabla_{R_i} \tilde{f}(\{t_i, R_i\}_{i\in[N]}) = 2\alpha (R_i - \bar{R}_i) + (1-\alpha) \sum_{j:(i,j)\in E} 2(t_j - t_i)(R_i^\top (t_j - t_i) - \tilde{t}_{ij})^\top.$$

Equivalently, we know

$$\nabla_{R(q_i)} \bar{f}(x) = 2\alpha (R(q_i) - \bar{R}_i) + (1-\alpha) \sum_{j:(i,j)\in E} 2(t_j - t_i)(R(q_i)^\top (t_j - t_i) - \tilde{t}_{ij})^\top.$$

Finally, we again use chain rule, according to

$$\nabla_{q_i} \bar{f}(x) = \begin{bmatrix} \langle \nabla_{R(q_i)} \bar{f}, \frac{\partial R(q_i)}{\partial q_{iw}} \rangle_F \\ \langle \nabla_{R(q_i)} \bar{f}, \frac{\partial R(q_i)}{\partial q_{ix}} \rangle_F \\ \langle \nabla_{R(q_i)} \bar{f}, \frac{\partial R(q_i)}{\partial q_{iy}} \rangle_F \\ \langle \nabla_{R(q_i)} \bar{f}, \frac{\partial R(q_i)}{\partial q_{iz}} \rangle_F \end{bmatrix}$$

with

$$\mathrm{grad}_{q_i} f = P_{q_i}(\nabla_{q_i} \bar{f}).$$

*E. Dictionary Learning*

Setup:

- Let $y_1, \ldots, y_k \in \mathbb{R}^d$ be a collection of $k$ images of dimension $d$
- Let $b_1, \ldots, b_n \in \mathbb{R}^d$ be a dictionary of $n$ atoms of dimension $d$
- Let $Y = \begin{bmatrix} y_1 & \cdots & y_k \end{bmatrix} \in \mathbb{R}^{d \times k}$ be the "data matrix"
- Let $D = \begin{bmatrix} b_1 & \cdots & b_n \end{bmatrix} \in \mathbb{R}^{d \times n}$ be the "dictionary matrix"
- We normalize $b_i$ such that $\|b_i\|_2^2 = 1$, meaning $b_i \in S^{d-1}$ and $D \in (S^{d-1})^n$
- Let $C \in \mathbb{R}^{n \times k}$ be the "coefficient matrix"
- We seek to find $D \in (S^{d-1})^n$ and $C \in \mathbb{R}^{n \times k}$ such that $Y \approx DC$ and $C$ is sparse
- For sparsity of $C$, we allow $n > d$
- For dictionary compression, we want $n << k$

Discontinuous problem:

$$\min_{D \in (S^{d-1})^n, C \in \mathbb{R}^{n \times k}} f(D, C) = \|Y - DC\|_F^2 + \lambda \|C\|_0$$

Continuous, non-smooth problem:

$$\begin{aligned} \min_{D \in (S^{d-1})^n, C \in \mathbb{R}^{n \times k}} f(D, C) &= \|Y - DC\|_F^2 + \lambda \|C\|_1 \\ &= \|Y - DC\|_F^2 + \lambda \sum_{j=1}^{n} \sum_{\ell=1}^{k} |C_{j\ell}| \end{aligned}$$

Smooth problem:

$$\min_{D \in (S^{d-1})^n, C \in \mathbb{R}^{n \times k}} f(D, C) = \|Y - DC\|_F^2 + \lambda \sum_{j=1}^{n} \sum_{\ell=1}^{k} \sqrt{C_{j\ell}^2 + \varepsilon}$$

We split $n$ into $N$ partitions $n_1, \ldots, n_N$ such that $\sum_{i=1}^{N} n_i = n$. Agent $i$'s decision variables will be $D_i \in (S^{d-1})^{n_i}$ and $C_i \in \mathbb{R}^{n_i \times k}$ such that

$$D = \begin{bmatrix} D_1 & \cdots & D_N \end{bmatrix}, \qquad \text{and} \qquad C = \begin{bmatrix} C_1 \\ \vdots \\ C_N \end{bmatrix}.$$

Agent $i$'s manifold is the product manifold

$$\mathcal{M}_i = (S^{d-1})^{n_i} \times \mathbb{R}^{n_i \times k}.$$

The problem can be restated as

$$\min_{D_i \in (S^{d-1})^{n_i}, C_i \in \mathbb{R}^{n_i \times k}} f\left(\{(D_i, C_i)\}_{i \in [N]}\right) = \left\| Y - \sum_{i=1}^{N} D_i C_i \right\|_F^2 + \lambda \sum_{i=1}^{N} \sum_{j=1}^{n_i} \sum_{\ell=1}^{k} \sqrt{(C_i)_{j\ell}^2 + \varepsilon}$$

Let $D_{ij} \in S^{d-1}$ be the $j^{\text{th}}$-column of $D_i$ and let $C_{ij} \in \mathbb{R}^{1 \times k}$ be the $j^{\text{th}}$-row of $C_i$, such that

$$D_i = \begin{bmatrix} D_{i1} & \cdots & D_{in_i} \end{bmatrix}, \qquad \text{and} \qquad C_i = \begin{bmatrix} C_{i1} \\ \vdots \\ C_{in_i} \end{bmatrix}.$$

The objective function can be restated as

$$f\left(\{(D_i, C_i)\}_{i \in [N]}\right) = \left\| Y - \sum_{i=1}^{N} \sum_{j=1}^{n_i} D_{ij} C_{ij} \right\|_F^2 + \lambda \sum_{i=1}^{N} \sum_{j=1}^{n_i} \sum_{\ell=1}^{k} \sqrt{(C_i)_{j\ell}^2 + \varepsilon}$$

Gradient with respect to $D_{IJ}$ for $I \in [N]$ and $J \in [n_I]$:

Objective

$$
\begin{aligned}
f\left(\{(D_i, C_i)\}_{i\in[N]}\right) &= \left\| Y - \sum_{i=1}^{N}\sum_{j=1}^{n_i} D_{ij}C_{ij} \right\|_F^2 + \lambda \sum_{i=1}^{N}\sum_{j=1}^{n_i}\sum_{\ell=1}^{k} \sqrt{(C_i)_{j\ell}^2 + \varepsilon} \\
&= \mathrm{Tr}\left( \left( Y - \sum_{i=1}^{N}\sum_{j=1}^{n_i} D_{ij}C_{ij} \right)^\top \left( Y - \sum_{i=1}^{N}\sum_{j=1}^{n_i} D_{ij}C_{ij} \right) \right) + \lambda \sum_{i=1}^{N}\sum_{j=1}^{n_i}\sum_{\ell=1}^{k} \sqrt{(C_i)_{j\ell}^2 + \varepsilon} \\
&= \mathrm{Tr}\left( Y^\top Y - 2Y^\top \sum_{i=1}^{N}\sum_{j=1}^{n_i} D_{ij}C_{ij} + \left( \sum_{i=1}^{N}\sum_{j=1}^{n_i} C_{ij}^\top D_{ij}^\top \right) \left( \sum_{i=1}^{N}\sum_{j=1}^{n_i} D_{ij}C_{ij} \right) \right) \\
&\quad + \lambda \sum_{i=1}^{N}\sum_{j=1}^{n_i}\sum_{\ell=1}^{k} \sqrt{(C_i)_{j\ell}^2 + \varepsilon} \\
&= \mathrm{Tr}\left(Y^\top Y\right) - 2\mathrm{Tr}\left( Y^\top \sum_{i=1}^{N}\sum_{j=1}^{n_i} D_{ij}C_{ij} \right) + \mathrm{Tr}\left( \left( \sum_{i_1=1}^{N}\sum_{j_1=1}^{n_{i_1}} C_{i_1 j_1}^\top D_{i_1 j_1}^\top \right) \left( \sum_{i_2=1}^{N}\sum_{j_2=1}^{n_{i_2}} D_{i_2 j_2}C_{i_2 j_2} \right) \right) \\
&\quad + \lambda \sum_{i=1}^{N}\sum_{j=1}^{n_i}\sum_{\ell=1}^{k} \sqrt{(C_i)_{j\ell}^2 + \varepsilon} \\
&= \mathrm{Tr}\left(Y^\top Y\right) - 2\mathrm{Tr}\left( \sum_{i=1}^{N}\sum_{j=1}^{n_i} Y^\top D_{ij}C_{ij} \right) + \mathrm{Tr}\left( \sum_{i_1=1}^{N}\sum_{j_1=1}^{n_{i_1}}\sum_{i_2=1}^{N}\sum_{j_2=1}^{n_{i_2}} C_{i_1 j_1}^\top D_{i_1 j_1}^\top D_{i_2 j_2}C_{i_2 j_2} \right) \\
&\quad + \lambda \sum_{i=1}^{N}\sum_{j=1}^{n_i}\sum_{\ell=1}^{k} \sqrt{(C_i)_{j\ell}^2 + \varepsilon} \\
&= \mathrm{Tr}\left(Y^\top Y\right) - 2\mathrm{Tr}\left( \sum_{i=1}^{N}\sum_{j=1}^{n_i} C_{ij}Y^\top D_{ij} \right) + \mathrm{Tr}\left( \sum_{i_1=1}^{N}\sum_{j_1=1}^{n_{i_1}}\sum_{i_2=1}^{N}\sum_{j_2=1}^{n_{i_2}} D_{i_1 j_1}^\top D_{i_2 j_2}C_{i_2 j_2}C_{i_1 j_1}^\top \right) \\
&\quad + \lambda \sum_{i=1}^{N}\sum_{j=1}^{n_i}\sum_{\ell=1}^{k} \sqrt{(C_i)_{j\ell}^2 + \varepsilon} \\
&= \mathrm{Tr}\left(Y^\top Y\right) - 2\mathrm{Tr}\left( \sum_{i=1}^{N}\sum_{j=1}^{n_i} D_{ij}^\top Y C_{ij}^\top \right) + \mathrm{Tr}\left( \sum_{i_1=1}^{N}\sum_{j_1=1}^{n_{i_1}}\sum_{i_2=1}^{N}\sum_{j_2=1}^{n_{i_2}} D_{i_1 j_1}^\top D_{i_2 j_2}C_{i_2 j_2}C_{i_1 j_1}^\top \right) \\
&\quad + \lambda \sum_{i=1}^{N}\sum_{j=1}^{n_i}\sum_{\ell=1}^{k} \sqrt{(C_i)_{j\ell}^2 + \varepsilon} \\
&= \mathrm{Tr}\left(Y^\top Y\right) - 2\sum_{i=1}^{N}\sum_{j=1}^{n_i} D_{ij}^\top Y C_{ij}^\top + \sum_{i_1=1}^{N}\sum_{j_1=1}^{n_{i_1}}\sum_{i_2=1}^{N}\sum_{j_2=1}^{n_{i_2}} D_{i_1 j_1}^\top D_{i_2 j_2}C_{i_2 j_2}C_{i_1 j_1}^\top \\
&\quad + \lambda \sum_{i=1}^{N}\sum_{j=1}^{n_i}\sum_{\ell=1}^{k} \sqrt{(C_i)_{j\ell}^2 + \varepsilon}
\end{aligned}
$$

Euclidean Gradient:

$$\begin{aligned}
\nabla_{D_{IJ}} f\left(\{(D_i, C_i)\}_{i\in[N]}\right) &= -2YC_{IJ}^\top + \sum_{i_2=1}^{N}\sum_{j_2=1}^{n_{i_2}} D_{i_2j_2}C_{i_2j_2}C_{IJ}^\top + \sum_{i_1=1}^{N}\sum_{j_1=1}^{n_{i_1}} D_{i_1j_1}C_{IJ}C_{i_1j_1}^\top \\
&= -2YC_{IJ}^\top + \sum_{i_2=1}^{N}\sum_{j_2=1}^{n_{i_2}} D_{i_2j_2}C_{i_2j_2}C_{IJ}^\top + \sum_{i_1=1}^{N}\sum_{j_1=1}^{n_{i_1}} D_{i_1j_1}C_{i_1j_1}C_{IJ}^\top \\
&= -2YC_{IJ}^\top + 2\sum_{i=1}^{N}\sum_{j=1}^{n_i} D_{ij}C_{ij}C_{IJ}^\top \\
&= -2YC_{IJ}^\top + 2DCC_{IJ}^\top \\
&= -2(Y - DC)C_{IJ}^\top \\
\nabla_{D_I} f\left(\{(D_i, C_i)\}_{i\in[N]}\right) &= -2(Y - DC)C_I^\top
\end{aligned}$$

Gradient with respect to $(C_I)_{JL}$ for $I \in [N]$, $J \in [n_I]$, and $L \in [k]$:
Objective:

$$\begin{aligned}
f\left(\{(D_i, C_i)\}_{i\in[N]}\right) &= \mathrm{Tr}\left(Y^\top Y\right) - 2\sum_{i=1}^{N}\sum_{j=1}^{n_i} D_{ij}^\top YC_{ij}^\top + \sum_{i_1=1}^{N}\sum_{j_1=1}^{n_{i_1}}\sum_{i_2=1}^{N}\sum_{j_2=1}^{n_{i_2}} D_{i_1j_1}^\top D_{i_2j_2}C_{i_2j_2}C_{i_1j_1}^\top \\
&\quad + \lambda\sum_{i=1}^{N}\sum_{j=1}^{n_i}\sum_{\ell=1}^{k}\sqrt{(C_i)_{j\ell}^2 + \varepsilon} \\
&= \mathrm{Tr}\left(Y^\top Y\right) - 2\sum_{i=1}^{N}\sum_{j=1}^{n_i}\sum_{\ell=1}^{k}(D_{ij}^\top Y)_{1\ell}(C_i)_{j\ell} + \sum_{i_1=1}^{N}\sum_{j_1=1}^{n_{i_1}}\sum_{i_2=1}^{N}\sum_{j_2=1}^{n_{i_2}}\sum_{\ell=1}^{k} D_{i_1j_1}^\top D_{i_2j_2}(C_{i_2})_{j_2\ell}(C_{i_1})_{j_1\ell} \\
&\quad + \lambda\sum_{i=1}^{N}\sum_{j=1}^{n_i}\sum_{\ell=1}^{k}\sqrt{(C_i)_{j\ell}^2 + \varepsilon}
\end{aligned}$$

Gradient:

$$\begin{aligned}
\nabla_{(C_I)_{JL}} f\left(\{(D_i, C_i)\}_{i\in[N]}\right) &= -2(D_{IJ}^\top Y)_{1L} + 2\sum_{i=1}^{N}\sum_{j=1}^{n_i} D_{IJ}^\top D_{ij}(C_i)_{jL} + \lambda\frac{(C_I)_{JL}}{\sqrt{(C_I)_{JL}^2 + \varepsilon}} \\
&= -2\left(D_{IJ}^\top(Y - DC)\right)_{1L} + \lambda\frac{(C_I)_{JL}}{\sqrt{(C_I)_{JL}^2 + \varepsilon}} \\
\nabla_{C_I} f\left(\{(D_i, C_i)\}_{i\in[N]}\right) &= -2D_I^\top(Y - DC) + \lambda C_I \oslash \sqrt{C_I \circ C_I + \mathbf{1}^{n_I\times k}\varepsilon},
\end{aligned}$$

where $\oslash$, $\circ$, and $\sqrt{\cdot}$ respectively denote the element-wise division, multiplication, and square root operators.